

Introduction

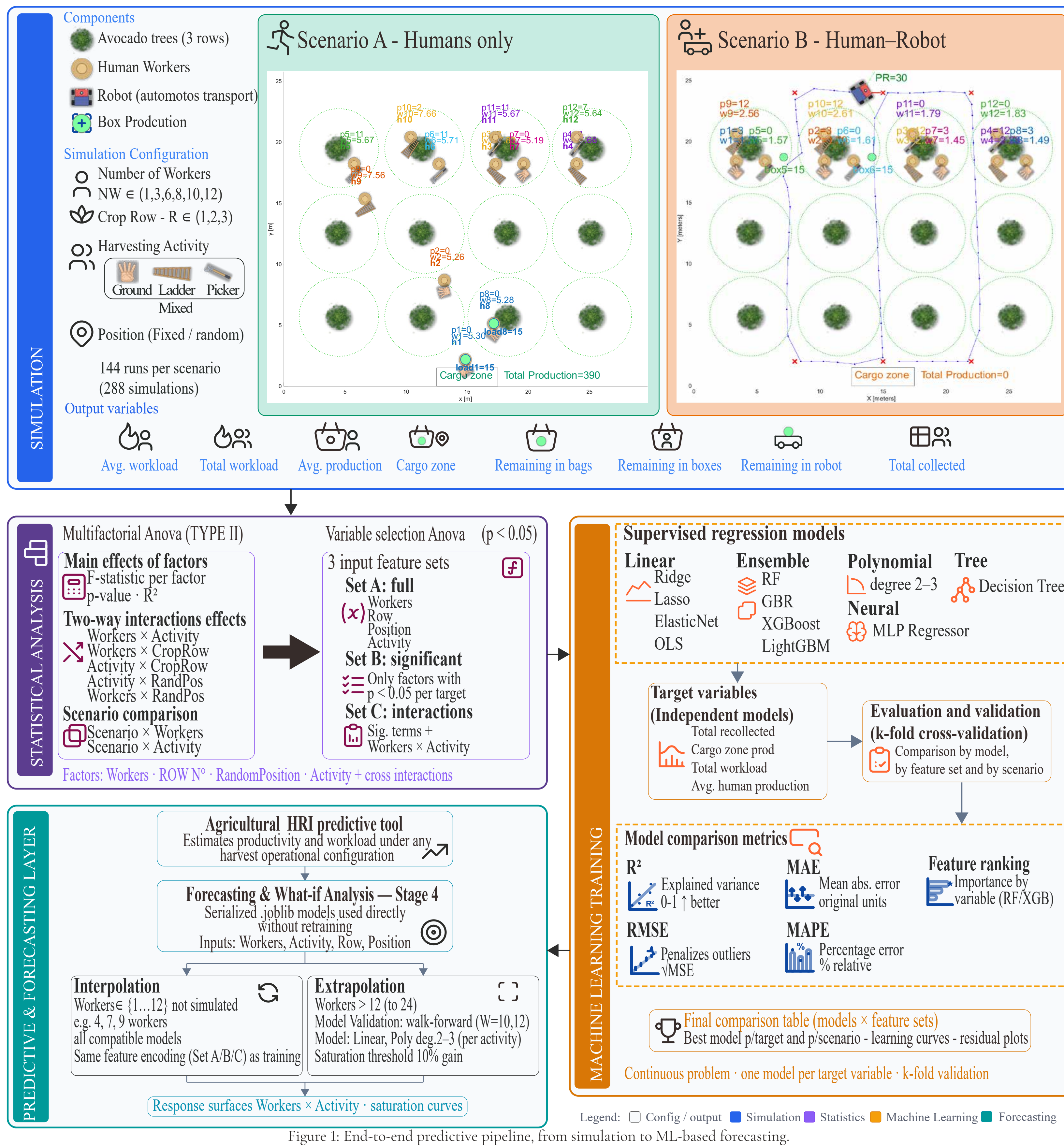
Human-robot collaboration (HRC) in agricultural harvesting is a promising but still emerging strategy for addressing farm labour shortages and improving orchard productivity, particularly in labour-intensive crops such as avocado. Agent-based simulators can evaluate specific HRC configurations in detail, but each new scenario requires a full simulation run, which limits their practical use for rapid, on-demand operational planning. This work extends a previously validated avocado-harvesting HRC simulator (Vásconez & Auat Cheein, 2022) with a supervised machine learning forecasting layer, transforming a scenario-by-scenario evaluation tool into a general-purpose predictive decision-support instrument capable of estimating production and human physical workload for configurations that were never explicitly simulated.

Objectives

- Extend the MATLAB-based HRC simulator (Vásconez & Auat Cheein, 2022) to support larger workforce sizes (up to 12 workers, vs. 6 in the original implementation), generating an expanded factorial dataset for model training.
- Train and validate machine learning models capable of predicting human physical workload and harvesting production for avocado-collection configurations that were never explicitly simulated.
- Compare model families and engineered feature sets to identify the most reliable predictors per target variable and scenario.
- Use the validated models to identify the workforce size at which a collaborative robot's contribution to production becomes operationally marginal.

Methodology

Two agent-based MATLAB simulators (Human-Only, Human-Robot) generated 288 runs varying workforce size, harvesting activity, crop row, and position; physical workload was estimated via ISO 8996:2004 metabolic coefficients. A Type II multifactorial ANOVA identified significant factors, informing three engineered feature sets (full, significant-only, and interaction-augmented). Twelve regression model families were trained per target variable and scenario, validated via 5-fold cross-validation and an independent hold-out split. The full pipeline is summarized in Fig. 1.



Results

The Human-Robot cooperative strategy increased total harvested production by 34-64% and cargo-zone delivery by 16-41% relative to the Human-Only scenario, while human physical workload increased by only 5-13% – confirming that robot assistance improves throughput without materially increasing operator strain. The best-performing models (gradient boosting and polynomial regression) achieved cross-validation $R^2 \geq 0.98$ for all four target variables (total recollected crops, cargo-zone production, total workload, and average production) across both scenarios, enabling sub-second point predictions for workforce configurations not explicitly simulated – a runtime reduction of several orders of magnitude relative to a full simulation run.

Robot Contribution Saturation

A robot-contribution saturation analysis identified the worker count at which the robot's productivity gain becomes operationally marginal (<10%). Since in-sample fit does not guarantee reliable extrapolation, the extrapolating model itself was chosen via a dedicated walk-forward validation: each candidate was refit using only data below the two highest training levels (Workers = 10, 12) and evaluated against the real, never-fitted value at that level. Using the validated extrapolant, no harvesting activity fell below the 10% saturation threshold within the training range, nor up to Workers = 24 (Fig. 2), indicating a wide operational margin before robot assistance stops being worthwhile.

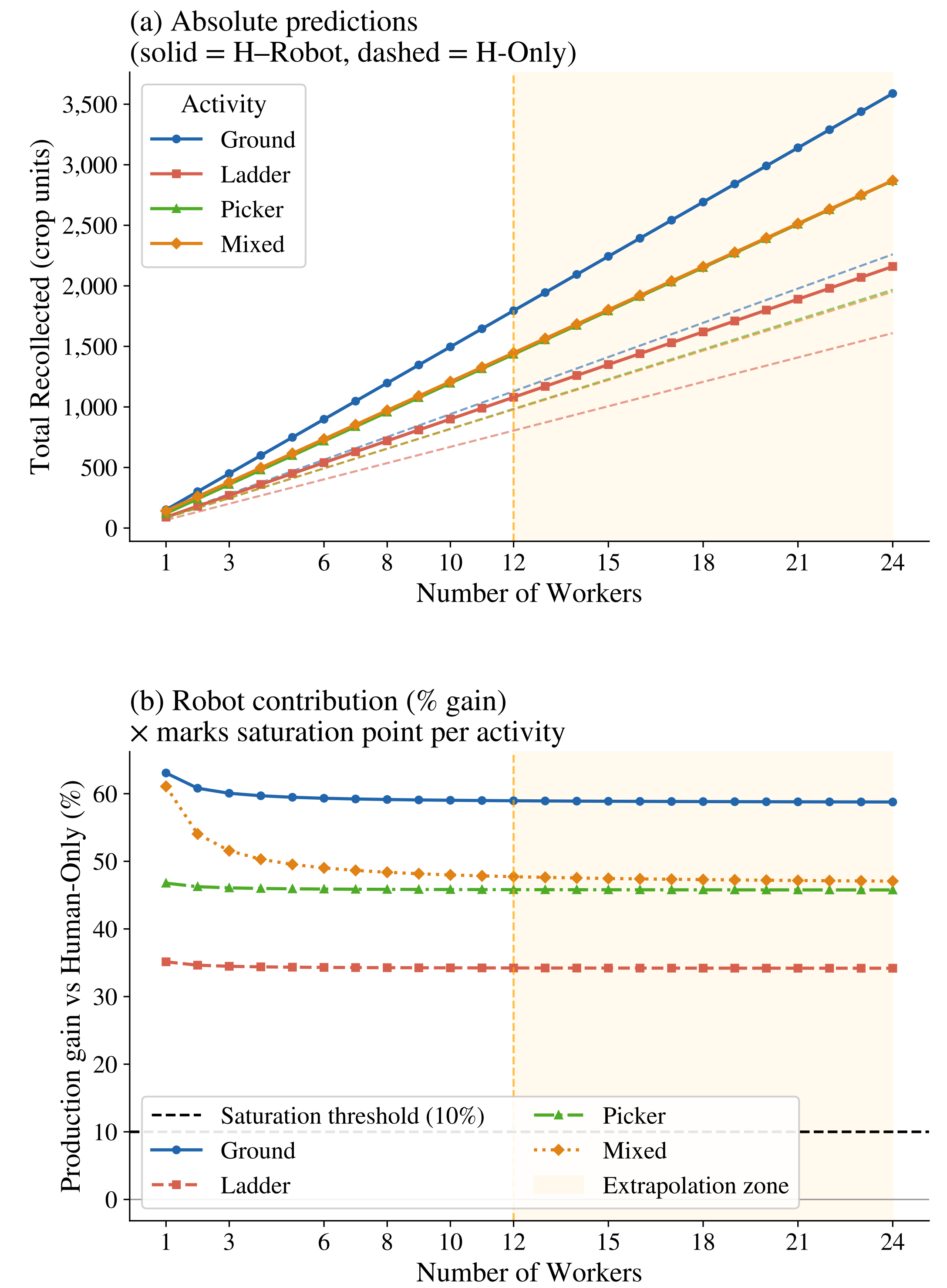


Figure 2: Robot-contribution saturation analysis for TotalRecollected (10% saturation threshold; held-out-validated model per activity).

Model Performance Summary

Scenario	Target	Best model	CV R^2	HO RMSE
H-Only	Tot. Recollected	Poly deg 3	0.999	12.16
H-Only	Cargo Zone	Poly deg 3	0.998	15.79
H-Only	Tot. Workload	Grad. Boosting	0.9996	1.35
H-Only	Avg. Production	Grad. Boosting	0.991	0.97
H-Robot	Tot. Recollected	Grad. Boosting	0.9998	2.44
H-Robot	Cargo Zone	Grad. Boosting	0.981	48.17
H-Robot	Tot. Workload	Grad. Boosting	1.000*	0.29
H-Robot	Avg. Production	Decision Tree	1.000*	0.00

* $R^2 = 1.000$ reflects the deterministic simulator, not real-world predictive power (see Limitations, main paper).

Conclusion

The proposed ANOVA-informed machine learning pipeline transforms an agent-based HRC simulator into a lightweight operational planning tool, enabling farm managers to estimate harvesting performance and physical workload under arbitrary workforce and task configurations without running additional simulations. The consistently high predictive accuracy across model families confirms that production and workload in this collaborative harvesting task are strongly structured by a small number of operational factors, supporting the use of this framework for orchard-level decision-making. Future work will extend the approach to other high-value tree crops and incorporate real-field sensor data to validate the simulator's estimates against measured outcomes.

References

- [1] Vásconez, J.P., Auat Cheein, F., 2022. Workload and production assessment in the avocado harvesting process using human-robot collaborative strategies. *Biosystems Engineering* 223, 56-77.
- [2] International Organization for Standardization, 2004. ISO 8996:2004 – Ergonomics of the thermal environment – Determination of metabolic rate.
- [3] Langsrud, Ø., 2003. ANOVA for unbalanced data: Use Type II instead of Type III sums of squares. *Statistics and Computing* 13(2), 163-167.
- [4] Pedregosa, F., et al., 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* 12, 2825-2830.
- [5] Duckett, T., et al., 2018. Agricultural robotics: the future of robotic agriculture. *arXiv preprint arXiv:1806.06762*.